MATH0112 Manifolds

Year:	2026–2027
Code:	MATH0112
Level:	7 (UG), 7 (PG)
Normal student group(s):	UG: Y4 Mathematics programmes PG: MSc Mathematics
Value:	15 credits (7.5 ECTS credits)
Term:	1
Assessment:	90% examination, 10% coursework
Normal Pre-requisites:	MATH0019 is recommended.
Lecturer:	Prof L Louder

Course Description and Objectives

Manifolds are a central concept in modern mathematics, they are the abstract concept of a 'smooth space', a general setting in which we can do differentiation and integration. They are fundamental in geometry and topology, and are used in many other areas of pure and applied maths and theoretical physics. The aim of this course is to give a thorough introduction to the theory of abstract smooth manifolds, including many examples.

Recommended Texts

- WM Boothby, *An Introduction to Differentiable Manifolds and Riemannian Geometry*
- M Spivak, *A Comprehensive Introduction to Differential Geometry* Volume 1

Detailed Syllabus

Core topics include:

Review of submanifolds in real vector spaces and the Implicit Function Theorem. Examples. Abstract definition of a smooth manifold, examples. Quotient constructions. Smooth maps. Immersions, submersions, diffeomorphisms. Tangent and cotangent spaces. Derivative of a smooth map. Degree of a smooth map, topological applications. Vector fields. Poincaré-Hopf theorem. Flows. Lie bracket.

Additional topics will be selected from:

Differential forms, Stokes' theorem. De Rham cohomology. Manifolds with boundary. Tangent and cotangent bundles, vector bundles. Riemannian metrics, curvature. Symplectic structures, complex structures.

September 2026